

Control the water level in a tank

Equipment

TwinCAT 3 and the simulation “Twincat project Tank”

Preparation

- Download the project ”Twincat project tank” from Canvas to your computer
- Click on the file ”TwinCAT Project Tank.sln” to launch the project
- When TC3 started, start and select run <UmRT_Default>
- Click on ”Activate Configuration”  and then answer OK to download the configuration to your runtime PLC.

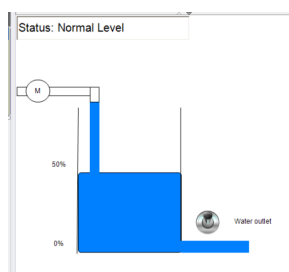
Programming Tasks

Task 1

Use the Function Block Diagram programming language that is prepared in PLCstudent.

Create an on/off control of the tank. For that, you need two comparison functions, LE and GE, which are equal to $\leq$ and $\geq$. Use the functions to compare the actual level variable “GVL.iwActualLevel”. The water level should not go below 40% or above 60%. Use the MOVE function block with EN to move or assign values to the “GVL.qwPumpSpeed” variable to set the speed of the incoming water. Set the speed to 40 if the water level is high, to 60 if the level is low, and to zero if the level is too high. Play a round with the water outlet rocker switch on and off for testing.

Insert text strings into the text variable “GVL.qsStatusText” to display the actual status. Use compare and move functions for that as well. The text could be ‘Low level’, ‘High level’, and ‘Normal level’



Task 2

Create a simple proportional P controller. The formula for a P controller is $(Set-Is)*K_p$. Let the Set be a constant of 50 and the Is variable the “GVL.iwActualLevel”. Start with a K_p of 10, increase it later until oscillation occurs. Use the SUB function to calculate the difference between Set and Is, and MUL for the amplification K_p of the difference.

OBS! The SUB function will not work correctly on WORD variables because they cannot be negative; instead, they will turn to a high positive value. Use the function WORD_TO_INT on the “GVL.iwActualLevel” before the SUB function and go back to WORD again with the INT_TO_WORD function before assigning to “GVL.qwPumpSpeed”.

You may need to declare integer variables between the different functionalities. You still have to handle the negative values by, for example, using the SEL function to set negative values to zero.

Task 3

Do tasks 1 and 2 in the Structured Text (ST) instead of FBD. Use $\geq$, $\leq$, -, *, :=, IF THEN END_IF instead of the FBD functions.

Hand in all four solutions to Canvas.